

Presentation 2

Link to Presentation 2 Slides

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Menu%20redundancy%20analysis/RepeatedProcesses_23Oct_WillVinceYiHyun.pdf

Link to Code

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Menu%20redundancy%20analysis/MainMenuRedundancies_23Oct_YiHyun.ipynb

What did you do?

Our group (Will, Vincent, Yi Hyun) recategorized the two types of redundancies that we defined in the last presentation into types of people with four subtypes. We've brainstormed the assumptions and algorithms that will divide the CAR data into these four types.

Once we agreed on the definitions, I first analyzed the Main Menu to identify the proportion of each type of caller. The results showed that the majority of people did not listen to the full prompt but were still able to navigate through the call without coming back to the main menu. My second analysis showed the distribution of the number of visits to the MainMenu prompt and found out the majority of calls did not have the caller returning back to the Main Menu. Finally, I focused on call sessions when the caller did have to return back to the Main Menu and found what menu they were coming from and the proportion of those. One significant finding was that the majority of callers proceeded to the LegalMenu2 after the MainMenu, but rarely had to come back to the MainMenu. This is an indication that the MainMenu prompt is concise and efficient enough for callers to jump from MainMenu to LegalMenu2.

How does it help the project?

These analyses are an extension of our definition of redundancies and help our objective of finding possible redundancies within the caller experience and if yes, why that is the case and what segments of the call the redundancies are coming from. In particular, my analyses this time helped our group shift our focus away from the MainMenu since it appeared to be less of a problem of confusion, both path and prompt wise.

Issues faced (if any)

Our main issue was that the CAR data did not tell us if the call was in English or Spanish. We've noticed that the prompt length of English and Spanish was quite different (10+ sec difference), usually Spanish prompts being longer. This made our time analysis on determining whether or not the caller listened to the full option not as accurate since we had to choose a duration that will roughly cover both durations. For example, for the Main Menu prompt analysis, the duration of the English prompt is 52 seconds while the Spanish prompt is 70 seconds so I had to consider all callers exceeding a call duration over 52 seconds to be categorized into "fully

listened to”. This unfortunately includes Spanish callers that did not listen to the full Spanish prompt which can affect our time analysis results.

Attempts to resolve issues (if any)

There was no way as of right now from the CAR data to find the language of the call.

Issues resolved (if any)

N/A

Next steps

Main Menu had a high percentage of people not listening to the full option and never returning back, which is a pretty good indication that Main Menu is not the primary segment causing confusion. Therefore, after finding out if multiple repetitions of Main Menu help callers navigate to a queue, we are planning to shift our focus on Legal Menu which seemed to have more confusion from the higher percentage of people returning back. In addition, there are much more sub-menus branching out for Legal Menu where we can analyze more granularly which paths are most frequent and which prompts might be confusing.

If the language is solvable, we also hope to analyze at which point of time during the prompt, people are leaving to analyze the impact of the duration of listening to a prompt has on the success rate of the caller reaching the right menu and also eventually reaching a queue. As of right now, we were only able to analyze at a binary level to “fully listened to” and “did not fully listen to”.

References

N/A

Presentation 1

Link to Presentation 1 Slides

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Menu%20redundancy%20analysis/RedundantPaths_09Oct_VincentWillYiHyun.pdf

What did you do?

Our group (Will, Vincent, Yi Hyun) performed flow chart analysis to brainstorm the different possible types and subtypes of each redundancy, the reasons why these should be defined as redundancies, and finally came up with potential algorithms in which those can be identified.

How does it help the project?

Each type of redundancy and proposed algorithm serves as a solid framework in which we can systematically categorize potential redundancies and eventually code to find segments of the call flow that have the most issues.

Issues faced (if any)

One issue as mentioned in the presentation is the potential discrepancy between the call flow chart and CAR data. This may become an issue since most of the proposed algorithms and subtypes were inspired from the flow chart itself and if the CAR data flows differently from the flow chart, our group may face difficulties backtracking each call path.

Attempts to resolve issues (if any)

We flagged one example of the discrepancy to the professor for confirmation/help.

Issues resolved (if any)

We are still waiting for confirmation/response from the professor

Next steps

My next step will involve focusing on a subtype of Type 2 redundancy to create the code that will replicate our proposed algorithm. This will eventually lead to knowing whether the type of subtype of redundancy is even a problem (observation of frequent occurrence) and if yes, if there are ways to mitigate that.

More specifically, our group is planning to analyze the CAR data to create a network chart or code that will map the flow for each caller id. This will help us find which nodes or edges most callers are stuck at or are being repeated.

The group will be collaborating across all types of redundancy. For example, I will be helping Vince by aggregating time data to find paths that take longer than they should or Will by further analyzing the menu chart for code redundancies.

References: N/A